

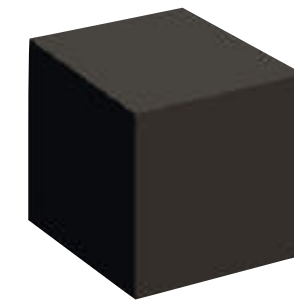
2. What is a computer?

1. Information Processing

A computer is a machine that has the function of **processing information**.

information processing

↑ function



Like any other machine, it has a definite function...

processing dough

↑ function



input



output

... but the function of a computer isn't something that can be described in purely physical terms.

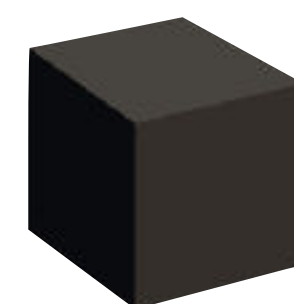
To process information is to perform **inferential transitions** between **contentful states**.

content

→ inferential transition

content

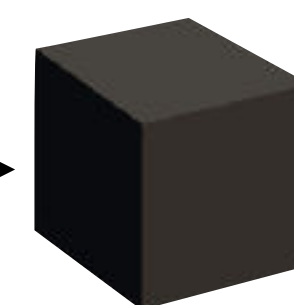
↑ function



2. Symbol Processing

A computer takes physical signals as **input** and produces physical signals as **output**.

symbol



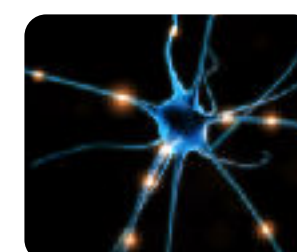
symbol

input

output

Symbols are physical objects or patterns. A computer takes symbols as inputs, processes symbols, and produces symbols as outputs.

apple

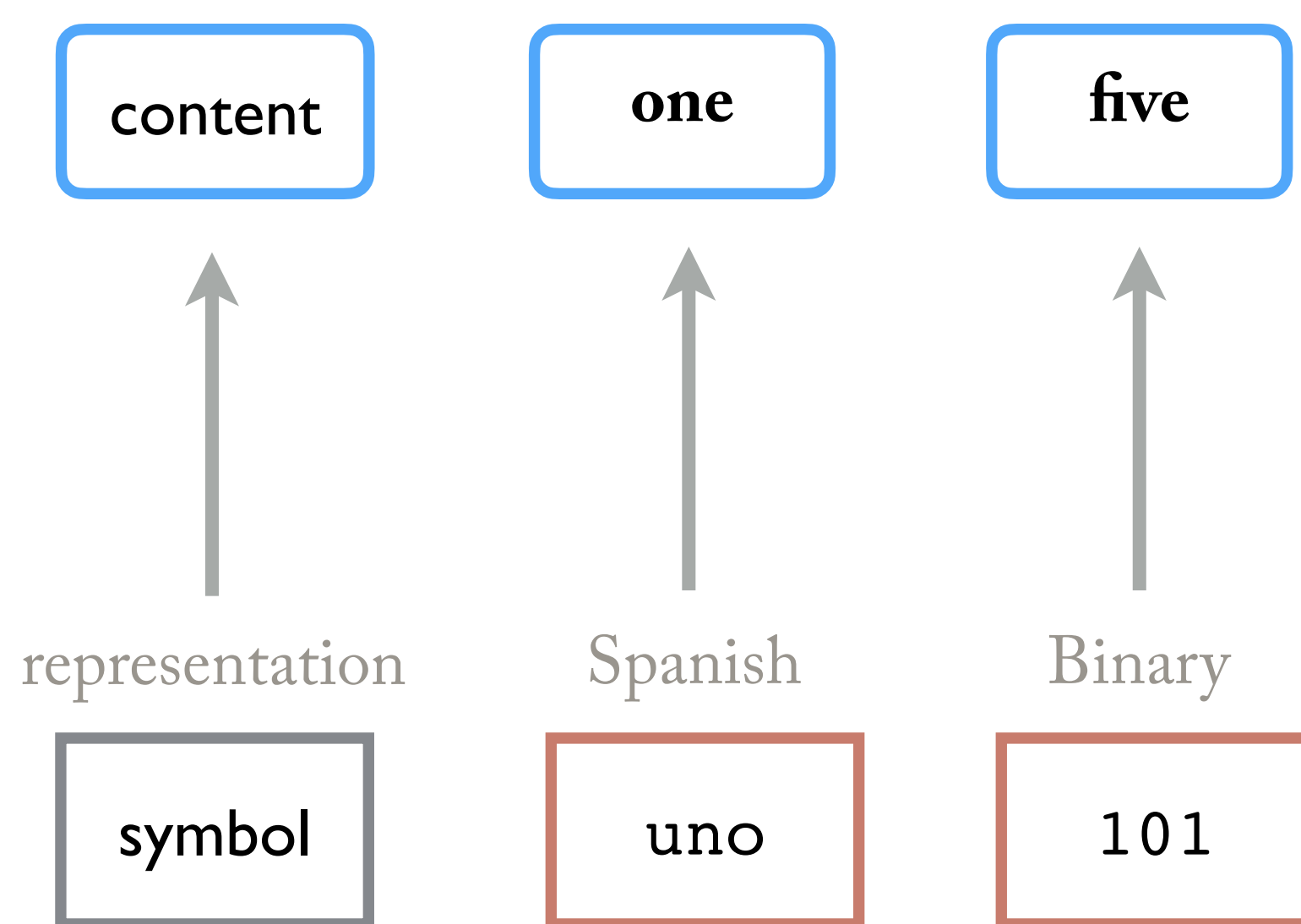


01010100
00100000
01110100
01110100

A **representation** is a physical symbol that is *about* something or expresses a meaning.

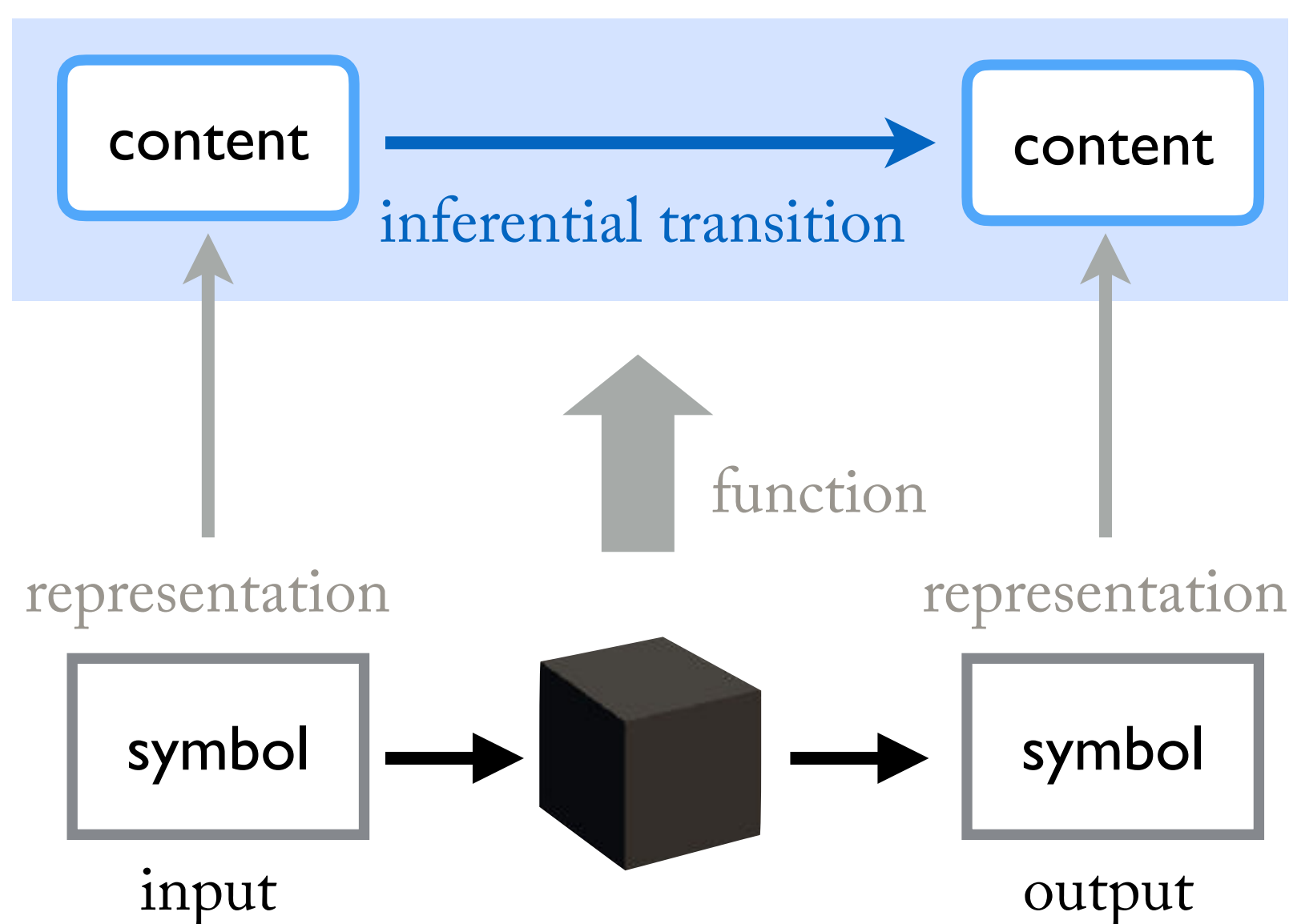
What a representation is about or what it means is its **content**. Symbols **represent** their contents.

Both **external** symbols, like words or pictures, and **internal** symbols, like computer code, can express content

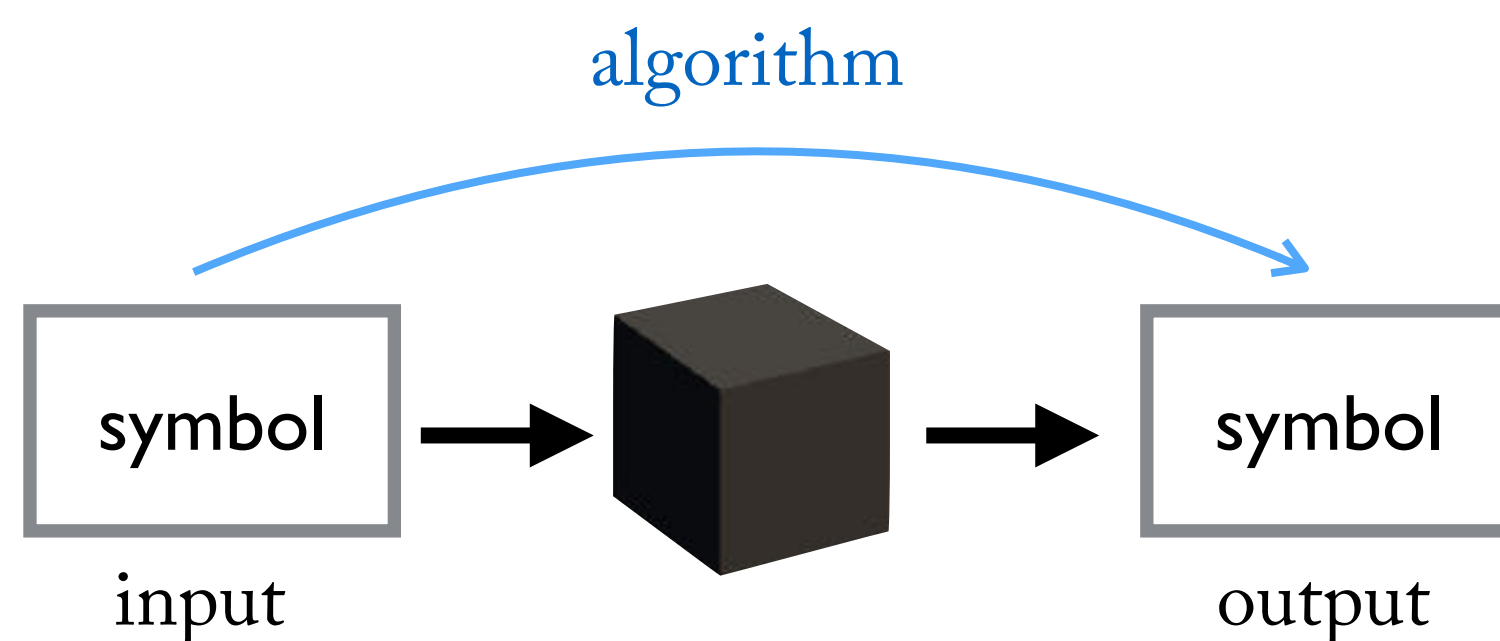


For a computer to perform an inferential transition, its inputs and outputs must represent the initial and final contents of the inferential transition.

The representations processed by the computer must **mirror** the inferential transition.



In order to produce representations which mirror inferential transitions, computers follow **algorithms**: mechanical procedures for manipulating symbols.

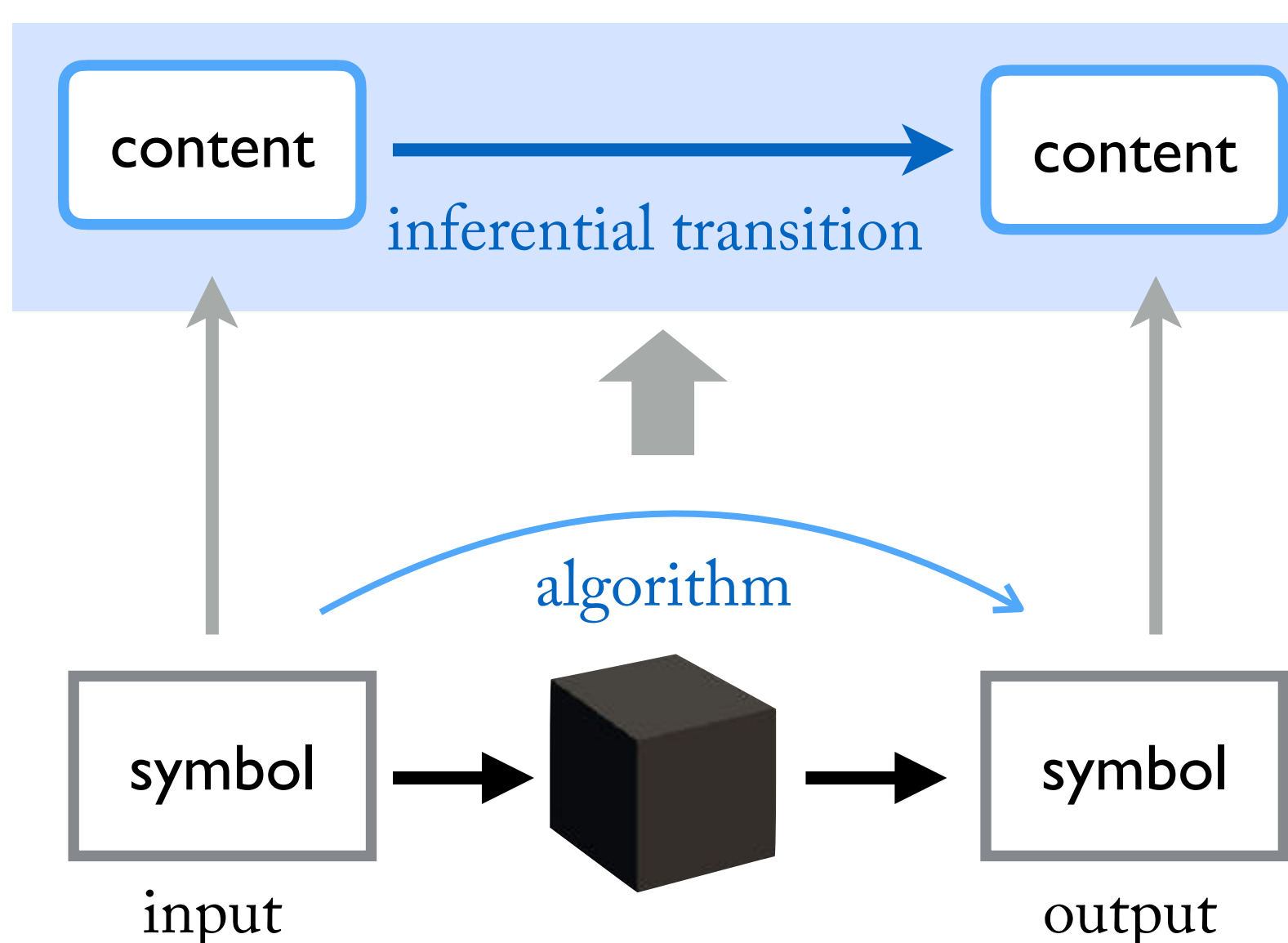


Computation

A computer processes information *by* processing symbols.

A computer functions to perform inferential operations by manipulating representational symbols according to an algorithm.

Computation as embodied logic.



Computational Theory of Mind (CTM)

Computational Theory of Mind (v. 1)

1. The brain is a naturally occurring computer or information processor.
2. Cognition arises from the computational activity of the brain.

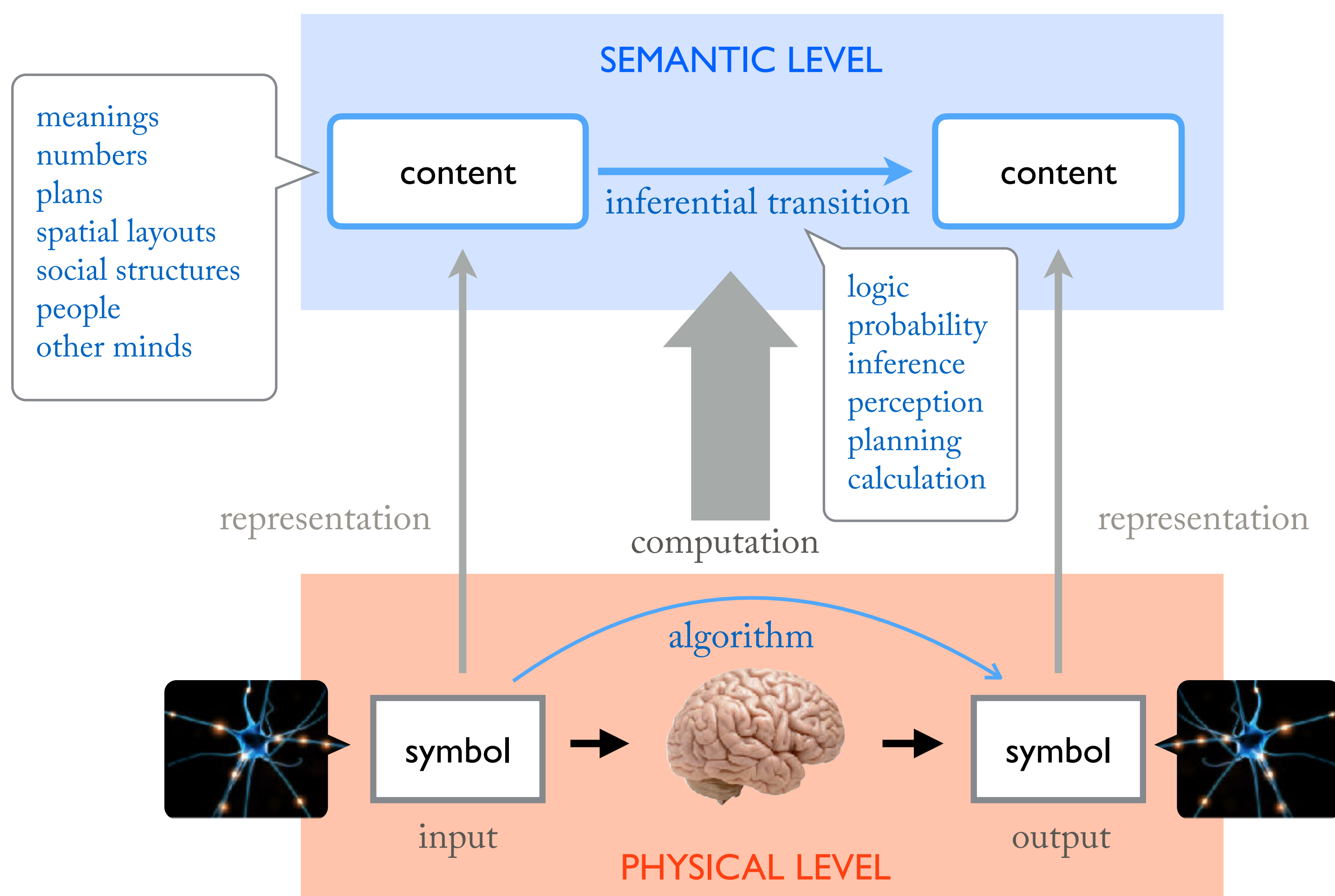
According to CTM, the brain is a type of computer.

Information processing

The brain has the function of processing information, or performing inferential operations. This is a biological function, not a design function.

Symbol processing

The brain takes symbols as inputs, it processes symbols, and produces symbols as outputs. Neural symbols in the brain represent mental contents.



Computational Theory of Mind: the brain computes cognitive functions by manipulating symbolic representations in the brain according to biologically grounded algorithms.

Further Notes

- How CTM answers Leibniz Challenge
Analogy with computers $\approx$ the brain is a computer, the mind is an algorithm.
- Computers are purely physical.
- They have thought-like states, and intelligence-like processes.
- They can cause and be caused by physical events.
- The physical domain and the semantic domain
- Natural and artificial computers.

- **CTM and Cognitive Science**
 - Cognitive science isn't philosophically committed.
 - But it relies on CTM as a working hypothesis.
 - CTM is a conjecture, not a conclusion.

- **Unit 1 outline:**
 - Logic circuits
 - Functions
 - Representation
 - Computation
 - CTM